

STM32 DEV BOARD

Overview

- A compact custom STM32 development board designed in KiCad, integrating USB-C connectivity, regulated 3.3 V power, reset/boot access, debug interfaces, GPIO headers and SPI/I²C/UART expansion.

Design item	Implemented design
Microcontroller	STM32F401RETx
Power	USB/VCC input with AMS1117-3.3 regulator
USB	USB-C receptacle with USB 2.0 D+/D- and CC circuitry
Clock	External crystal Y1 with associated capacitors
Interfaces	SWD, UART, SPI, I ² C and GPIO headers
PCB validation	DRC: 0 violations; 0 unconnected items
Schematic validation	ERC: 0 violations; 0 warnings

1. Problem Statement

The project addresses the need for a compact, reusable STM32F401-based development platform that can be programmed, debugged and connected to external electronics without requiring a separate breakout board. The design combines the microcontroller, power conditioning, USB-C interface, clock source, reset/boot controls and accessible signal headers on a single PCB.

2. Proposed Solution

A custom four-layer-capable-style compact PCB layout was developed in KiCad around the STM32F401RETx MCU. The schematic includes an AMS1117-3.3 linear regulator, local bypass capacitors, ferrite-bead power isolation, an external crystal, reset circuitry, USB-C connectivity and grouped headers for debug and peripheral signals. The resulting board is mechanically rounded at the corners and includes four mounting holes.

3. Circuit Architecture

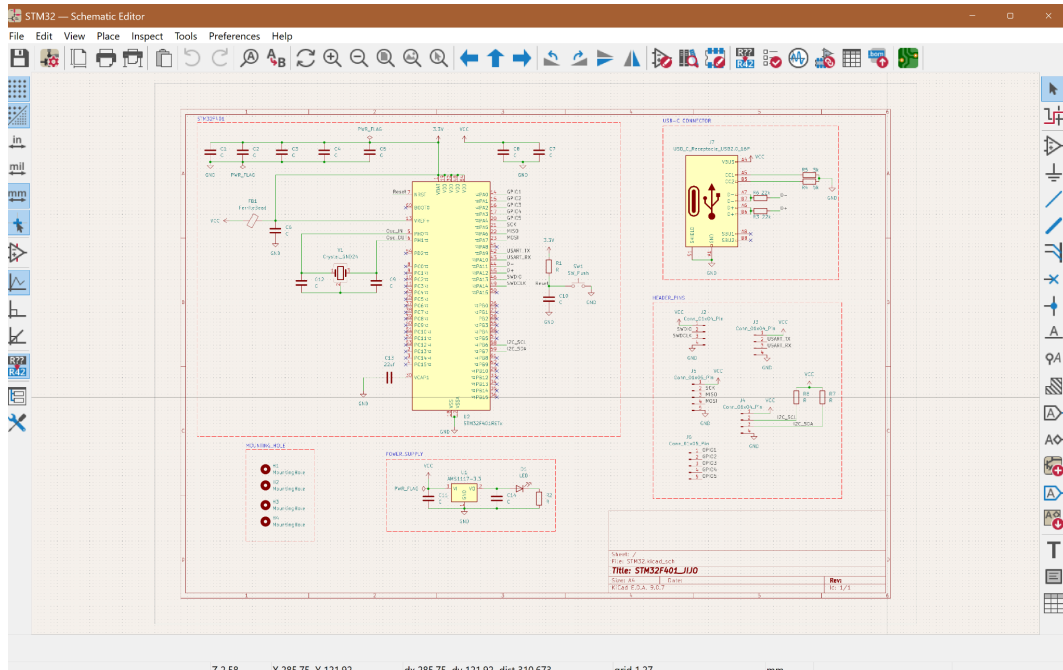


Figure 1. KiCad schematic showing the major circuit blocks.

4. PCB Design

The PCB layout places the STM32 MCU near the center, with the USB-C connector at the lower-right edge, power regulation toward the lower-left, and headers distributed around the board perimeter. This arrangement provides direct physical access to programming, communication and GPIO signals while keeping the main controller and decoupling network compact.

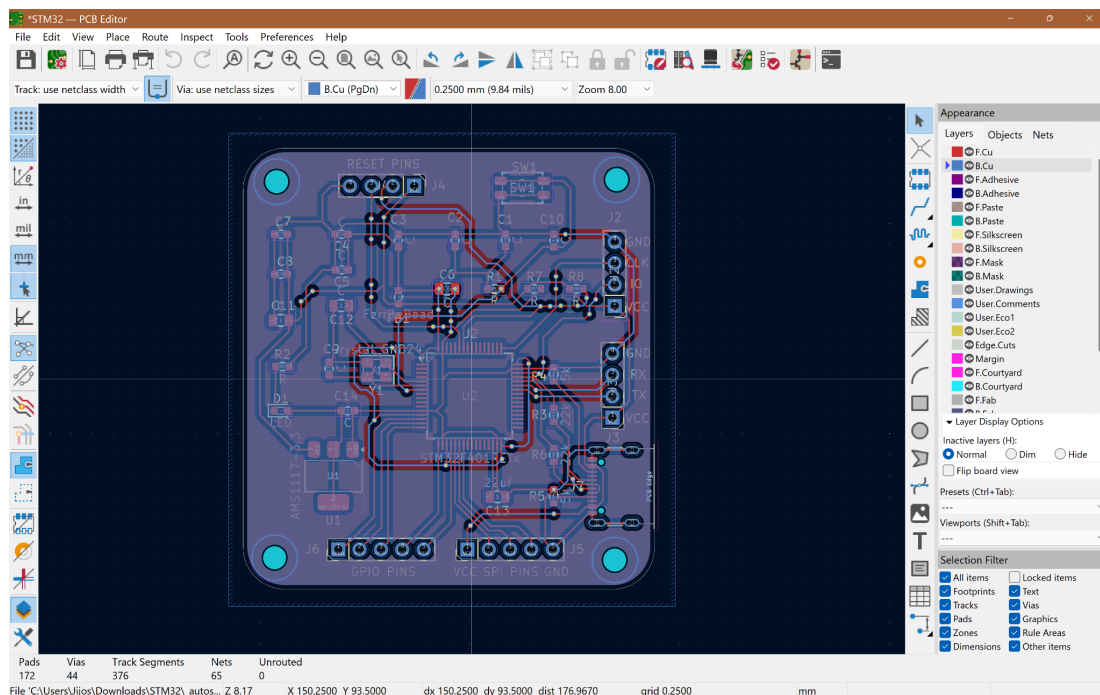


Figure 2. PCB Editor view with routing, zones

5. Major Components

Reference	Component / function
U2	STM32F401RETx — main 32-bit microcontroller
U1	AMS1117-3.3 — 3.3 V linear regulator
J7	USB-C USB 2.0 receptacle
Y1	External crystal clock source
SW1	Push-button reset/control input
FB1	Ferrite bead for supply filtering
J2–J6	GPIO, UART, SPI, I ² C and debug/header access
C1–C14	Supply, decoupling and clock-associated capacitors
D1	LED indicator
H1–H4	PCB mounting holes

Component	Ref.	Qty.	Est. Unit Cost (₹)	Est. Total (₹)
STM32F401RETx MCU	U2	1	350	350
AMS1117-3.3 Regulator	U1	1	15	15
USB-C Connector	J7	1	25	25
Crystal	Y1	1	10	10
Ferrite Bead	FB1	1	5	5
Capacitors	C1–C14	14	2	28
Resistors	R1–R8	8	2	16
LED	D1	1	3	3
Reset Push Button	SW1	1	5	5
Pin Headers	J2–J6	5 sets	15	75
Mounting Hardware	H1–H4	4	3	12
Estimated Prototype Cost				₹844

6. Design Considerations

- Local decoupling capacitors are distributed around the MCU and power circuitry.
- The ferrite bead separates/filters the incoming supply path before the MCU rail.
- USB-C, SWD and peripheral signals are routed to accessible board-edge headers/connectors.
- Four mounting holes support mechanical integration into a fixture or enclosure.
- The layout uses a continuous board outline with rounded corners and a compact component footprint.
- Silkscreen reference designators and interface labels are used to improve assembly and bring-up.

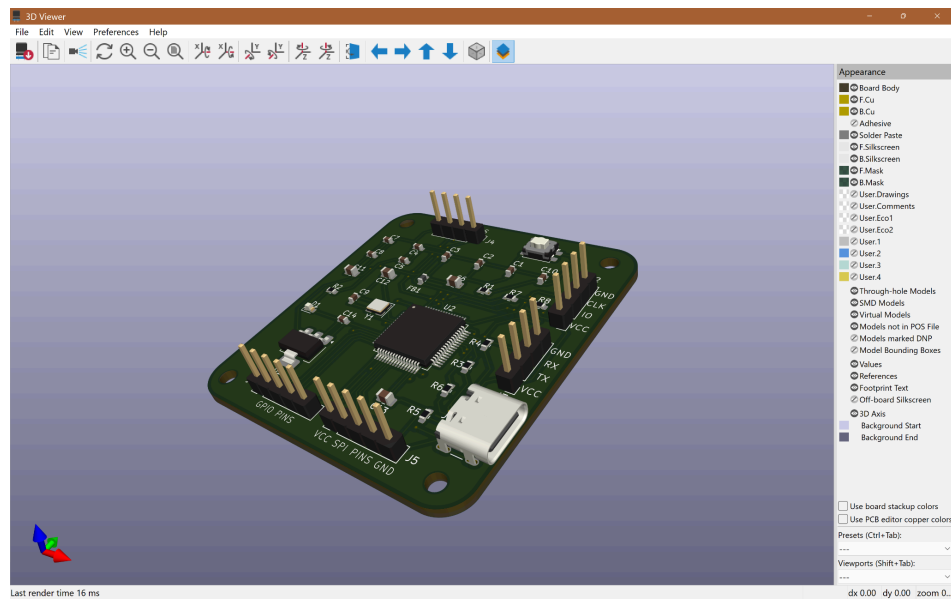


Figure 3. 3D view

7. Testing and Validation

The supplied KiCad screenshots show successful rule-check results. In the PCB Design Rules Checker, the reported counts are 0 violations and 0 unconnected items. The schematic Electrical Rules Checker also reports 0 violations and 0 warnings. These results provide evidence that the submitted design passes the configured KiCad DRC/ERC checks.

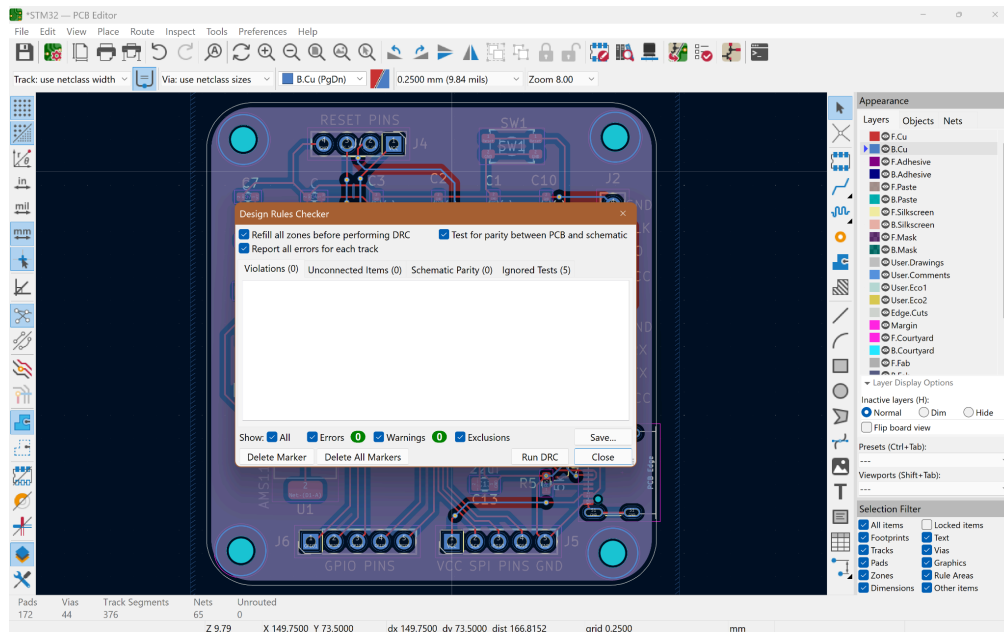


Figure 5. PCB Design Rules Checker: 0 violations and 0 unconnected items.

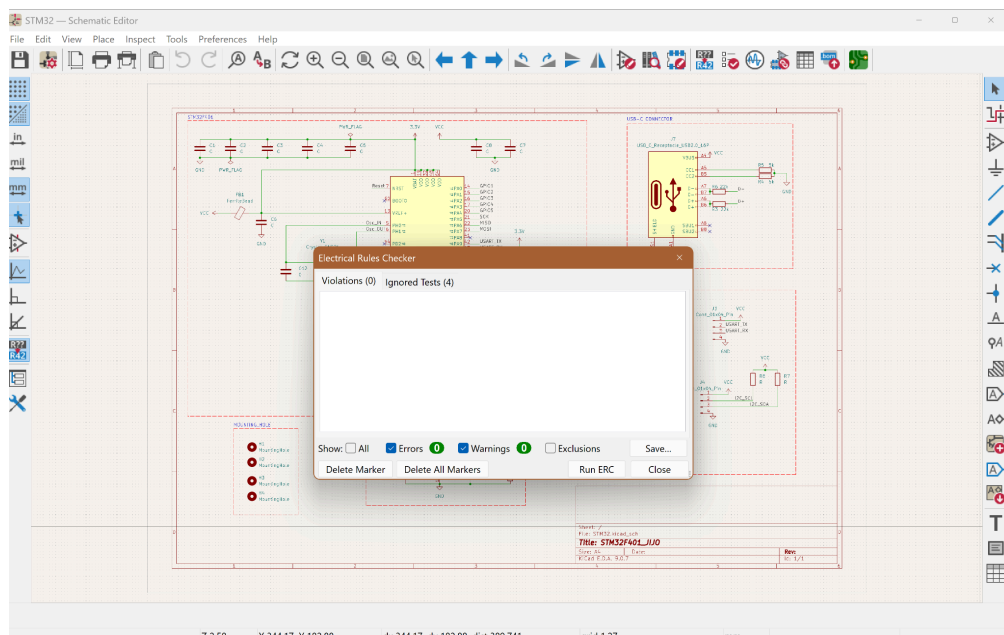


Figure 6. Electrical Rules Checker: 0 violations and 0 warnings.

The available evidence is CAD-level validation. No oscilloscope captures, electrical load tests, USB enumeration logs, thermal measurements or firmware test results were supplied, so those hardware-level results are not claimed in this report.

8. Challenges

The main design challenges are fitting the STM32 controller and supporting circuitry into a compact board, maintaining accessible connector placement, routing a dense set of MCU signals, and keeping power, clock and USB circuitry organized. Mechanical constraints such as the four mounting holes and edge-mounted USB-C connector also influence component placement and routing.

9. Final Outcome

The final design is a compact STM32F401RETx development board with USB-C, regulated 3.3 V power, external clocking, reset/boot access, debug connectivity and multiple peripheral/GPIO headers. The supplied PCB 3D views show a coherent assembled-board representation, while the reported KiCad DRC/ERC results show zero reported violations in the provided checks.

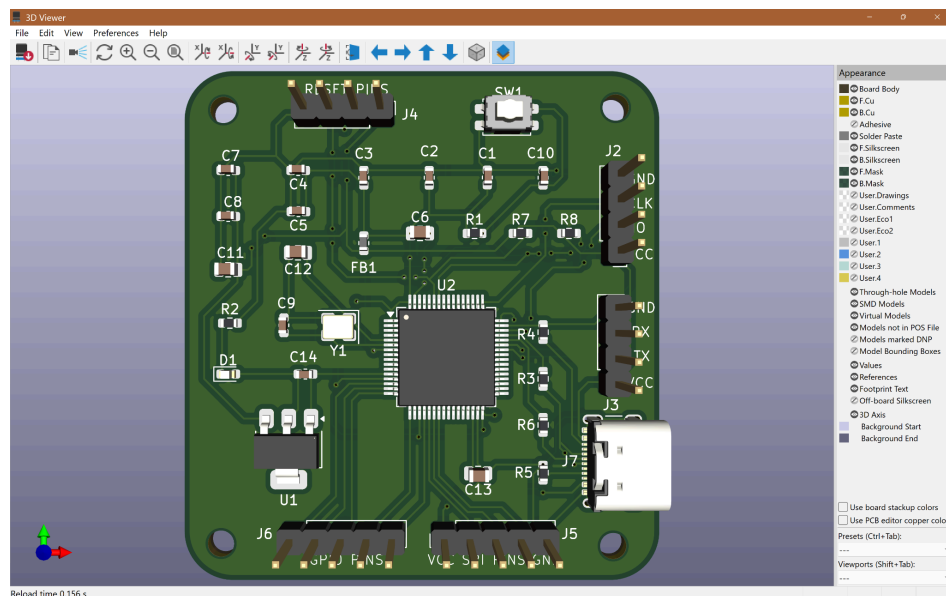


Figure 7. Final top-side 3D representation.

Conclusion:

The design is suitable as a compact STM32 development platform and provides a strong basis for prototype fabrication and firmware bring-up. Before production release, hardware-level validation and a final manufacturing review should be completed.